

Activity Dossier

Lab 1 — Rail Dataset Analysis

AI & Data Science with GenAI (Rail) · Your first data-analysis lab

Student package — everything you need to do this activity

Field	Detail
Module	Python / Data Analysis foundations
Builds on	Your pandas + describe(), IQR outliers, and matplotlib/seaborn sessions
Duration	~3 hours
You need	A browser + JupyterLab (or Google Colab)
Data	Real French TGV punctuality records (12,181 route-months, 2018–2026)
Hand in	Your completed notebook + this activity's workbook

1. The situation

You have joined the data team at a railway operator. You are handed a real, official dataset: the monthly punctuality record of French high-speed (TGV) trains. Every row is one route in one month — how many trains were scheduled, how many were cancelled, and how late they ran.

Your job today is to explore it — the same way you explored the machining dataset in class — and come back with a few clear, honest findings. This is exploratory data analysis (EDA): the first thing a data scientist does with any new dataset.

2. What you will be able to do

- Load a real, messy-formatted CSV into pandas and tidy up its columns.
- Summarise data with describe(), value_counts() and nunique().
- Find outliers with the IQR method and read a boxplot.
- Draw histograms, scatter plots and a correlation heatmap, and read them.
- Use groupby to compare groups — your one new tool this lab.

3. What's in your package

File	What it is
lab1_rail_dataset_analysis.ipynb	The notebook you work in — open this.

File	What it is
sncf_tgv_regularity.csv	The real dataset (already provided).
Student_Workbook_Lab1.docx	Where you write your answers and findings.
Activity_Dossier_Lab1.pdf	This document — your guide to the activity.
START_HERE.txt	One-page quick start.

4. Before you start

Open JupyterLab (or upload the notebook and the CSV to Google Colab). Keep the notebook and the CSV file in the same folder, then open the notebook and run cells from the top with Shift+Enter.

```
# In JupyterLab: open a terminal in the folder, then
jupyter lab
# then open lab1_rail_dataset_analysis.ipynb and run each cell top to bottom
```

How the notebook works: every step is fully worked and explained — just read and run it. After each step there is a small □ Your turn cell for you to practise, and it ends with an open □ Challenge. The steps get gradually harder; only the last one (groupby) is new.

5. The activity, step by step

Step	What you do	Already learned / New?
0–1	Load the real data and look at it (shape, info, describe)	Already learned
2	Count categories (value_counts, nunique) + a bar chart	Already learned
3	Distribution of one number (histogram + skewness)	Already learned
4	Find outliers with the IQR method (boxplot)	Already learned
5	Relationship between two numbers (scatter + correlation)	Already learned
6	All relationships at once (correlation heatmap)	Already learned
7	Compare groups with groupby (bar chart)	NEW — groupby
□	Challenge: two short analyses you run yourself	Your choice

6. What to hand in

Deliverable	Notes
Completed notebook	All steps run; the □ Your turn cells filled; the □ Challenge attempted.
Student Workbook	Findings recorded as you go, plus three findings and a recommendation.
Show & tell	Be ready to share one chart and one finding.

7. How you are assessed (100 marks)

Component	Marks	Full marks means
Worked steps run + □ Your turn cells done	40	Notebook runs top to bottom; practice cells completed
□ Challenge (both parts)	20	Both challenge parts answered with working code and a finding
Three findings + recommendation (workbook)	30	Correct, specific, in plain language
Charts are readable	10	Titles, axis labels and units on your charts

8. Your data — where it comes from

The dataset is real and official: SNCF Open Data, “Régularité mensuelle TGV” (monthly TGV punctuality). It is released under the Licence Ouverte / Open Licence (Etalab) — free to use, including commercially, with attribution. It is provided in your package, so you do not need to download anything.

The file is exactly as published, so its column names are in French. The notebook renames the ones you use to plain English (below). The data itself is untouched.

Column in file (French)	Renamed to	Meaning
date	month	Year-month of the record
service	service	National or International
gare_depart	from_station	Departure station
gare_arrivee	to_station	Arrival station
duree_moyenne	journey_time_min	Average journey time (minutes)
nb_train_prevu	scheduled_trains	Trains scheduled that month
nb_annulation	cancellations	Trains cancelled
nb_train_retard_arrivee	trains_late	Trains late on arrival
retard_moyen_arrivee	avg_delay_min	Average delay of the late arrivals (min)
nb_train_retard_sup_15	trains_over_15min_late	Trains more than 15 minutes late

9. Getting unstuck

- Run cells from the top in order — later cells depend on earlier ones.
- If loading fails, check the file is in the same folder and you used `sep=";"`.
- An outlier is not always an error — here, cancellation spikes are real strike months.
- A chart without a title and axis units is not finished.
- Stuck? Re-read the step's Insight box, then ask a teaching assistant.